Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Dicello, John F Ph.D.		
Project Title:	Radiation Effects: Core Project		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	diceljo@jhmi.edu	**Fax:**	FY 410-502-1419
PI Organization Type:	UNIVERSITY	**Phone:**	410-614-4194
Organization Name:	Johns Hopkins Cancer Center		
PI Address 1:	Harry & Jeanette Weinberg Bldg., Ste. 1440, Rm. 1454		
PI Address 2:	401 North Broadway		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21231-2410	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	01/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huso, David (Johns Hopkins University School of Medicine)		
Grant/Contract No.:	RE.002.03		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Approximately 3000 female Sprague-Dawley rats were sham irradiated or irradiated with energetic heavy ions in conjunction with the NASA program of the Brookhaven National Laboratory, with protons as part of the NASA program at Loma Linda University Medical Center, and with photons at Johns Hopkins School of Medicine. The animals are monitored for diseases, especially mammary carcinomas. Slightly less than half of the animals were subsequently treated with Tamoxifen to determine if this drug could effectively reduce the risk of breast cancer arising from the radiations.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	To experimentally determine carcinogenesis and other diseases in vivo resulting from low-dose irradiations with energetic heavy ions, protons, and gamma rays in order to ascertain the risks for astronauts in low-Earth orbit or long-duration missions in space. To determine if pharmaceutical intervention after irradiation can be used to reduce the risk of cancers.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 10/10/2006)
Articles in Peer-reviewed Journals	Dicello JF, Christian A, Cucinotta FA, Gridley DS, Kathirithamby R, Mann J, Markham AR, Moyers MF, Novak GR, Piantadosi S, Ricart-Arbona R, Simonson DM, Strandberg JD, Vazquez M, Williams JR, Zhang Y, Zhou H, Huso D. "In vivo mammary tumourigenesis in the Sprague-Dawley rat and microdosimetric correlates." Phys Med Biol. 2004 Aug 21;49(16):3817-30. PMID: 15446807 , Aug-2004
Articles in Peer-reviewed Journals	Karim BO, Landolfi JA, Christian A, Ricart-Arbona R, Qiu W, McAlonis M, Eyabi PO, Khan KA, Dicello JF, Mann JF, Huso DL. "Estrous cycle and ovarian changes in a rat mammary carcinogenesis model after irradiation, tamoxifen chemoprevention, and aging." Comp Med. 2003 Oct;53(5):532-8. PMID: 14655997 , Oct-2003